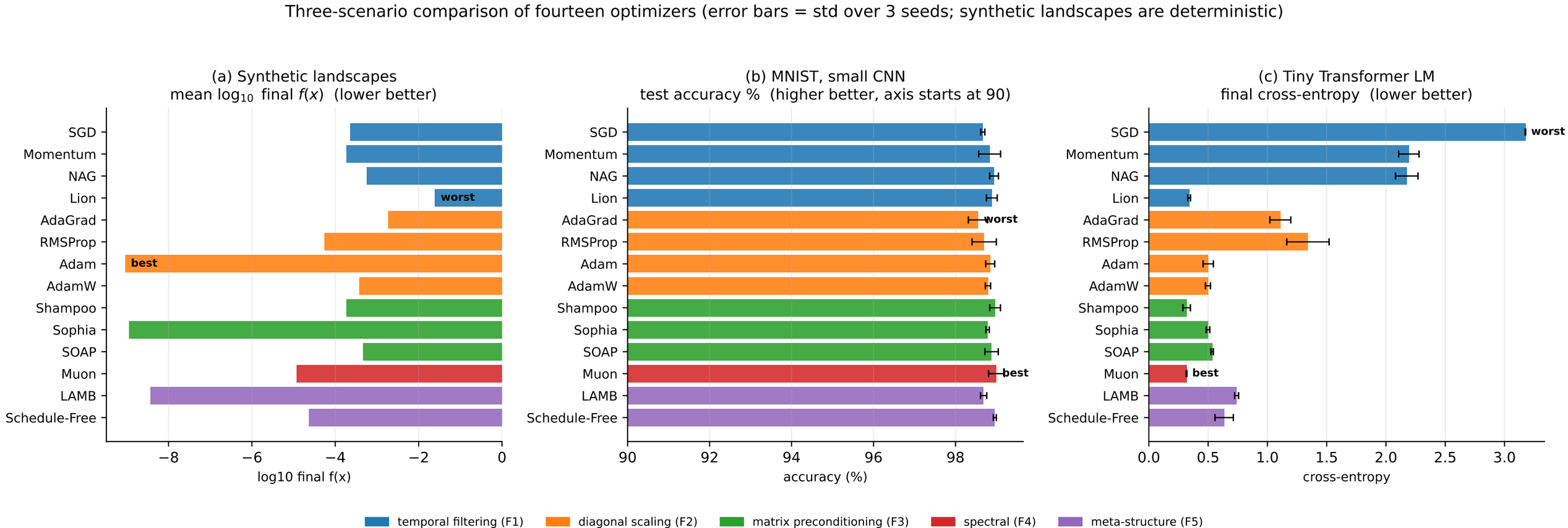
Three-scenario comparison of fourteen optimizers (error bars = std over 3 seeds; synthetic landscapes are deterministic)
(a) Synthetic landscapes
mean log10 final f(x) (lower better)
(b) MNIST, small CNN
test accuracy % (higher better, axis starts at 90)
(c) Tiny Transformer LM
final cross-entropy (lower better)
SGD
Momentum
NAG
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
Schedule-Free
worst
best
−8
−6
−4
−2
0
log10 final f(x)
90
92
94
96
98
accuracy (%)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
cross-entropy
temporal filtering (F1)
diagonal scaling (F2)
matrix preconditioning (F3)
spectral (F4)
meta-structure (F5)